

ED_Project Standard Deveints

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Aim and objectives

The aim of the experiment is to investigate whether the student's faculty of study affects the time required to solve a standardized logical reasoning test, comparing mean completion times across faculties using One-Way ANOVA. The research question is : Does the faculty of study influence the average time taken by Stellenbosch University students to solve a standardized logical reasoning test? This is worth investigating because if the mean completion time differs per faculty than we can proceed to say that faculties differ in the quantitative and abstract-reasoning content of their curricula. Engineering and Science students are trained in structured problem-solving under time-pressure while EMS students have less exposure to this structured reasoning style. If these curricular differences translate into measurable time differences on a logic test, then time-limited testing may disadvantage students from certain faculties regardless of their underlying reasoning ability. The specific comparison across Engineering, Science, and EMS at Stellenbosch University has not, to our knowledge, been previously examined, making this a novel context-specific test of a broader hypothesis. The results matters because the findings could inform how logical reasoning assessments are timed and applied across faculties. Also, the results contribute evidence on whether academic background functions as a confounding variable in timed cognitive assessments. The following steps will be undertaken:

- Recruit 10 consenting student volunteers from each of three faculties(EMS,Engineering,Science) for a total sample of $n = 30$.
- Make all the 30 students take the same standardized logical reasoning test under controlled conditions(no calculators,phones.AI tools,or internet access).
- Record each participant's total completion time in minutes.
- Summarise the data descriptively per faculty, supported by boxplots comparing the three groups.
- Test the assumptions underlying one-way ANOVA- normality and homoscedasticity of variance across faculties.
- Conduct one-way ANOVA to formally test for differences in mean completion time between faculties.

The statistical hypothesis for population mean completion times for Engineering, Science, and EMS students respectively.

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

$$H_1 : \text{At least one } \mu_i \text{ differs}$$

These hypotheses will be evaluated at a significance level of **0.05**

Experimental design

Experimental Units The experimental units are individual undergraduate students currently enrolled at Stellenbosch University

Factor(Independent Variable) Faculty of Study

Treatment Levels

Engineering Science Economic and Management Sciences (EMS)

Response Variable

The total time (in seconds) required to complete a standardised logical reasoning test.

Replication

Ten students will be recruited from each faculty, giving a total sample size of thirty(30) participants.

Experimental Procedure Each participant will complete exactly the same logical reasoning test under identical testing conditions. The timer will start once the participant begins the test and stop immediately after the final answer has been completed. The total completion time, measured in seconds, will be recorded as the response variable. Participants will complete the the assessment individually without assistance from other participants or access to calculators, mobile phones, artificial intelligence tools, or the internet.

Controlled Variables **1.** Test instrument - all participants will complete the identical standardised logical reasoning test, that is, same items, same order, same difficulty level. **2.** Academic year/level - all participants will be restricted to a similar year of study to reduce bias from academic maturity. **3.** Test administration - all questions will be typed out to reduce inconsistency. **4.** Time of the day - testing sessions will be around the same time of day to control fatigue or alertness effects. **5.** Resources available - participants will complete the the assessment individually without assistance from other participants or access to calculators, mobile phones, artificial intelligence tools, or the internet.

Data analysis

```
library(readxl)
library(hms) #library for removing the date
datat <- read_excel("Data.xlsx")
datat$Time_clean <- as_hms(datat$Time) #showing only the time in the data not the date
```

```
#The 5 number summary for each group
library(dplyr)
```

```
datat %>%
  group_by(Faculty) %>%
  summarise(average = mean(Time_clean),
            sd = sd(Time_clean),
            min = min(Time_clean),
            max = max(Time_clean),
            med = median(Time_clean)
  )
```

```
## # A tibble: 3 x 6
##   Faculty      average      sd min      max      med
##   <chr>      <drtn>    <dbl> <drtn>    <drtn>    <drtn>
## 1 EMS        29988 secs 16492. 11940 secs 64200 secs 27870 secs
## 2 ENGINEERING 27018 secs  9214. 14760 secs 43260 secs 25560 secs
## 3 SCIENCE    25692 secs 10043. 14040 secs 42300 secs 24000 secs
```

```
#Normality within each group
tapply(as.numeric(datat$Time_clean), datat$Faculty, shapiro.test)
```

```
## $EMS
##
## Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.92051, p-value = 0.3612
##
##
## $ENGINEERING
##
## Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.95911, p-value = 0.7757
##
##
## $SCIENCE
##
## Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.89221, p-value = 0.1796
```

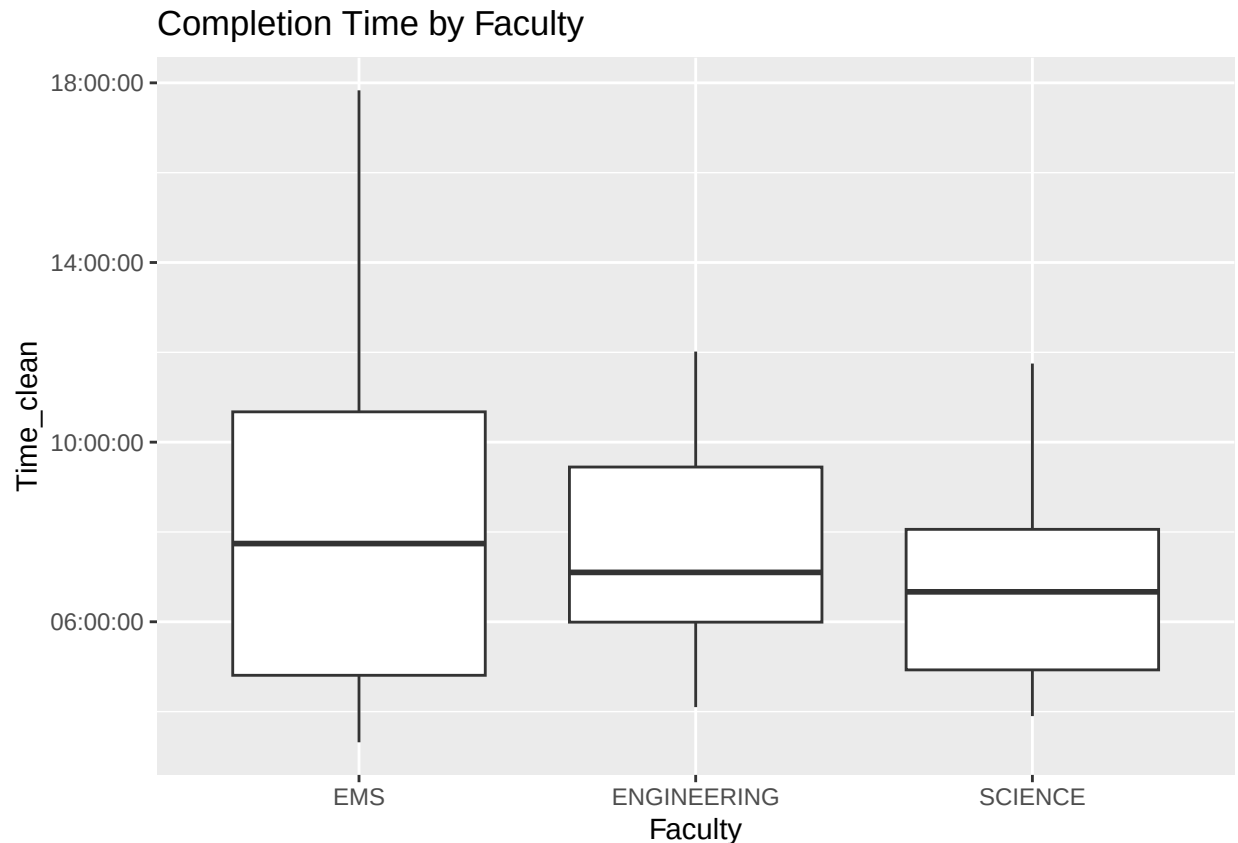
The normality test yields a $p = 0.3612$ for EMS, $p = 0.7757$ for Engineering, and $p = 0.1796$, which are all greater than 0.05. Therefore normality is not rejected for any group.

```
#Homoscedasticity
bartlett.test(Time_clean ~ Faculty, data = datat)
```

```
##
## Bartlett test of homogeneity of variances
##
## data:  Time_clean by Faculty
## Bartlett's K-squared = 3.62, df = 2, p-value = 0.1637
```

The homoscedasticity test yields a $p = 0.1637$ which is greater than 0.05. We can therefore not reject it, because this indicates that the assumptions of variances is not rejected.

```
#Boxplots for each faculty
library(ggplot2)
ggplot(datat,aes(x = Faculty, y = Time_clean))+geom_boxplot()+
  labs(title = "Completion Time by Faculty")
```



```
#ANOVA model
model_datat <- aov(Time_clean ~ Faculty, data = datat)
summary(model_datat)
```

```
##           Df      Sum Sq   Mean Sq F value Pr(>F)
## Faculty    2 9.678e+07  48391320   0.317  0.731
## Residuals 27 4.120e+09 152581240
```

The ANOVA yields a $p = 0.731$ which is greater than 0.05. We fail to reject the null hypothesis, concluding that the faculty of study does not have a significant effect on the time it takes to solve logical problems. All of the assumptions are valid for our statistical test regarding the ANOVA, therefore we can interpret the results with confident.

Main findings

A One-Way ANOVA was conducted to test whether does the faculty(EMS,Engineering,Science) of study affect the time it takes for Stellenbosch University students to solve logic based questions. At 5% significant level, this results are not statistically significant as we fail to reject null hypothesis. There is insufficient evidence from the data to conclude that the faculty of study does affect the time it takes to solve logical problems. What we could have done differently is by choosing a larger sample size, as 10 per faculty limits the statistical power. We could have done more than 3 faculties, as others might have shown a different pattern from what we have. The assumptions of normality and homoscedasticity were assessed prior to interpreting the ANOVA results. Since both assumptions were satisfied, then we can confidently interpret the F-test results.